

2025 JUNE PURE MATHEMATICS PAPER 1 SOLUTIONS

6042/1 ADVANCED LEVEL

PURE MATHEMATICS PAPER 1 6042/1

3 hours

ONLINE REVISION

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INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the spaces provided.
 - Answer all questions.
 - Numerical answers should be given to appropriate accuracy.
-

INFORMATION FOR CANDIDATES

- The number of marks is given in brackets [] at the end of each question.
-

QUESTIONS

1. Find the series expansion of the function,

$$f(x) = \frac{3-x}{3+2x}$$

up to and including the term in x^3 . State the range of values of x for which the expansion is valid. [6]

2. Solve the equation:

$$\frac{3^x(9^x)^2}{81} = 27^x.$$

[4]

3. A graph has the following parametric equations:

$$x = \frac{\sin \theta}{\cos \theta}; \quad y = 1 - \sin^2 \theta; \quad \text{for } -\pi < 2\theta < \pi.$$

- (a) Find $\frac{dy}{dx}$. [4]

- (b) Find the coordinates of the turning point of the graph. [4]

4. Find the range of values of k for which the equation $kx^2 + 4x + k = 3$ has real roots. [5]

5. Variables x and y are related by the equation:

$$y = x^m e^n, \quad \text{where } m \text{ and } n \text{ are constants.}$$

- (a) Show that this relationship can be reduced to a linear form. [2]

- (b) A straight-line graph representing the relationship between x and y in (a) passes through points P and Q with coordinates $(30.5, 0.05)$ and $(0, -3)$, respectively. Find the value of m and the value of n . [3]

6. When the function $f(x) = (x-2)(x+3) + ax + b$ is divided by $x-2$, the remainder is 8, and when divided by $x+3$, the remainder is -2 .

- (a) Find the value of a and the value of b . [4]

- (b) Hence or otherwise, find the remainder when $f(x)$ is divided by $(x-2)(x+3)$. [2]

7. Express y in terms of x given that:

$$x \frac{dy}{dx} = y + 1, \quad \text{and } y = 4 \text{ when } x = 2.$$

[6]

8. A chord AB of a circle of radius 8 cm subtends an angle of $\frac{\pi}{8}$ radians at the centre of the circle. Calculate the:

(a) length of the major arc AB , [2]

(b) area enclosed by the chord AB and the minor arc AB . [4]

9. (a) Given:

$$P = \begin{pmatrix} -1 & 1 & 3 \\ 2 & -2 & -6 \\ -4 & 4 & 12 \end{pmatrix} \text{ and } Q = \begin{pmatrix} 3 & -2 & 0 \\ -1 & 3 & 7 \\ 2 & -1 & 1 \end{pmatrix},$$

find PQ . [3]

- (b) Show that:

$$\begin{pmatrix} -6 & -2 & 2 \\ 4 & -6 & 0 \\ 8 & 10 & -4 \end{pmatrix},$$

does not have an inverse. [3]

10. F varies directly as the square of W , and when $W = 830$, $F = 165$.

(a) Find the equation connecting F and W . [4]

(b) Calculate W when $F = 132$. [2]

11. (a) Show by calculation that the equation $e^{-\frac{1}{2}x} = 3x - 1$ has a root between $x = 0$ and $x = 1$. [3]

(b) Show that the equation $e^{-\frac{1}{2}x} = 3x - 1$ can be expressed as the iteration formula

$$x_{n+1} = -2 \ln(3x_n - 1)$$

[2]

(c) Taking $x_1 = 0.5$ and using the iteration formula

$$x_{n+1} = -2 \ln(3x_n - 1),$$

find the value of x_2 correct to 3 decimal places. [2]

12. Find $\sqrt[3]{1 - \sqrt{3}i}$ in the form $a + bi$. [8]

13. A line l passes through the point $(-2, 3)$ and is perpendicular to the line $2x - 3y = 4$.

(a) Find the coordinates of the point of intersection of l and the line $2x - 3y = 4$. [6]

(b) Find the distance between the point $(-2, 3)$ and the point of intersection of l and the line $2x - 3y = 4$, leaving the answer in exact form. [2]

14. It is given that $a * b = a + b + 1$ is a binary operation on the set of integers $\mathbb{Z}$.

(a) Find $-2 * 7$. [1]

(b) Show that the set of integers ($\mathbb{Z}$) under the binary operation $a * b = a + b + 1$ forms an Abelian group. [8]

15. The functions f and g are defined by:

$$f : x \rightarrow 3x + p, \quad g : x \rightarrow q - 2x; \quad x \in \mathbb{R},$$

where p and q are constants. Given that $ff(2) = 10$ and $g^{-1}(2) = 3$, find:

(a) the value of p and the value of q . [7]

(b) an expression for $fg(x)$ using the values of p and q from (a). [2]

16. Prove by induction that:

$$\frac{5^{2n} + 5}{5},$$

is divisible by 3, for all integer values of n . [10]

17. (a) Express:

$$f(x) = \frac{3x + 14}{(x - 2)(x^2 + 6)},$$

in the form:

$$\frac{A}{x - 2} + \frac{Bx + C}{x^2 + 6}.$$

[5]

(b) Hence or otherwise, find the exact value of:

$$\int_3^4 f(x) \, dx.$$

[6]

QUESTIONS

Question 1

$$\begin{aligned}\frac{3-x}{3+2x} &= (3-x)(3+2x)^{-1} \\ &= (3-x) \cdot 3^{-1} \left(1 + \frac{2}{3}x\right)^{-1} \\ &= \frac{1}{3}(3-x) \left[1 + (-1) \left(\frac{2}{3}x\right) + \frac{(-1)(-2)}{2!} \left(\frac{2}{3}x\right)^2 \right. \\ &\quad \left. + \frac{(-1)(-2)(-3)}{3!} \left(\frac{2}{3}x\right)^3 + \dots \right] \\ &= \left(1 - \frac{x}{3}\right) \left(1 - \frac{2}{3}x + \frac{4}{9}x^2 - \frac{8}{27}x^3 + \dots\right) \\ &= 1 - \frac{2}{3}x + \frac{4}{9}x^2 - \frac{8}{27}x^3 - \frac{x}{3} + \frac{2}{9}x^2 - \frac{4}{27}x^3 + \dots \\ &= 1 - x + \frac{2}{3}x^2 - \frac{4}{9}x^3 + \dots\end{aligned}$$

For validity;

$$-\frac{3}{2} < x < \frac{3}{2}$$

Question 2

$$\begin{aligned}\frac{3^x(9^x)^2}{81} &= 27^x \\ \frac{3^x(3^{2x})^2}{3^4} &= 3^{3x} \\ \frac{3^{x+4x}}{3^4} &= 3^{3x} \\ 5x - 4 &= 3x \\ 5x - 3x &= 4 \\ 2x &= 4 \\ x &= 2\end{aligned}$$

Question 3

(a)

$$\begin{aligned} x &= \frac{\sin \theta}{\cos \theta} \quad (\tan \theta) \\ \frac{dx}{d\theta} &= \frac{\cos \theta \cdot \cos \theta - \sin \theta (-\sin \theta)}{\cos^2 \theta} \\ &= \frac{\cos^2 \theta + \sin^2 \theta}{\cos^2 \theta} \\ &= \frac{1}{\cos^2 \theta} = \sec^2 \theta \end{aligned}$$

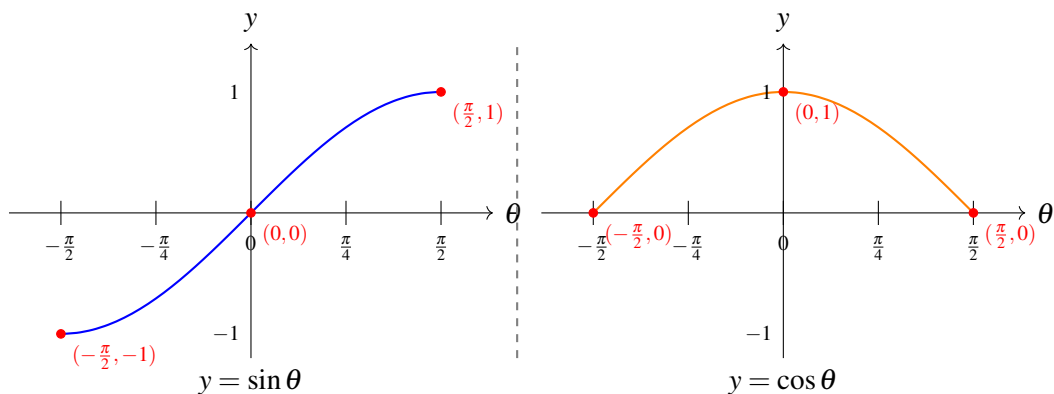
$$\begin{aligned} y &= 1 - \sin^2 \theta \quad (\text{since } \cos^2 \theta = 1 - \sin^2 \theta) \\ &= \cos^2 \theta \end{aligned}$$

$$\begin{aligned} \frac{dy}{d\theta} &= 2 \cos \theta \cdot (-\sin \theta) \\ &= -2 \sin \theta \cos \theta \\ &= -\sin 2\theta \end{aligned}$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{-2 \sin \theta \cos \theta}{\sec^2 \theta} \\ &= -2 \sin \theta \cos^3 \theta \quad \text{or} \quad \frac{-\sin 2\theta}{\sec^2 \theta} \end{aligned}$$

(b) For critical points ($dy/dx = 0$)

$$\begin{aligned} -2 \sin \theta \cos^3 \theta &= 0 \\ \Rightarrow -2 \sin \theta &= 0 \quad \text{or} \quad \cos^3 \theta = 0 \\ \sin \theta &= 0 \quad \text{or} \quad \cos \theta = 0 \\ \theta &= 0 \quad \text{or} \quad \theta = \pm \frac{\pi}{2} \quad \text{Note: Ignore since } -\frac{\pi}{2} < \theta < \frac{\pi}{2} \end{aligned}$$



If $\theta = 0$:

$$x = \tan(0) = 0$$

$$y = (\cos 0)^2 = 1$$

Therefore the coordinates = $(0; 1)$

Question 4

Given coordinates: $(0, 1)$

$$\text{Equation: } kx^2 + 4x + k = 3$$

$$kx^2 + 4x + (k - 3) = 0$$

For real roots ($b^2 - 4ac \geq 0$):

$$4^2 - 4(k)(k - 3) \geq 0$$

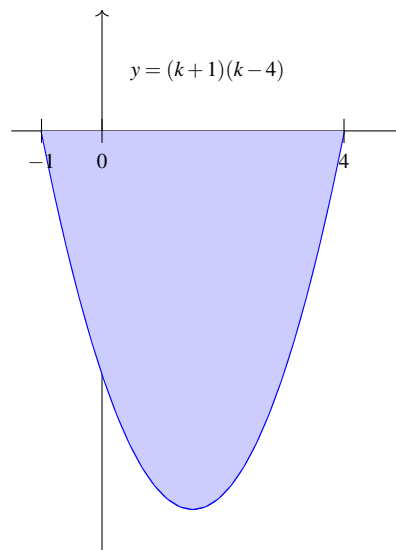
$$16 - 4k^2 + 12k \geq 0$$

$$4k^2 - 12k - 16 \leq 0$$

$$k^2 - 3k - 4 \leq 0$$

$$(k - 4)(k + 1) \leq 0$$

Critical values: $k = 4$ or $k = -1$



$$\therefore -1 \leq k \leq 4$$

Question 5

(a)

$$\begin{aligned}y &= x^m e^n \\ \ln y &= \ln(x^m e^n) \\ \ln y &= \ln x^m + \ln e^n \\ \ln y &= m \ln x + n\end{aligned}$$

In standard linear form:

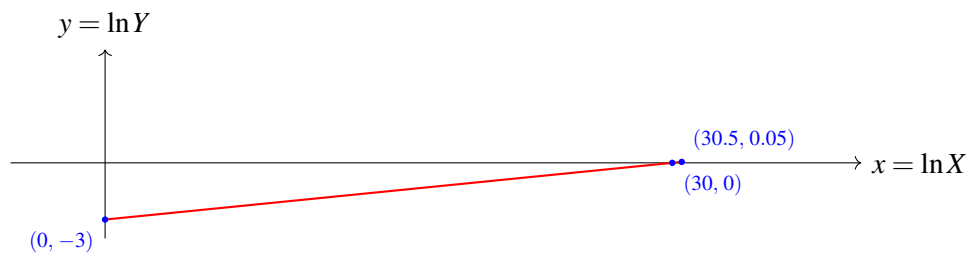
$$\ln y = m \ln x + n$$

which corresponds to:

$$Y = mX + c$$

where $Y = \ln y$, $X = \ln x$, and $c = n$.

(b) Sketch



$$\begin{aligned}m &= \frac{-3 - 0.05}{0 - 30.5} \\ m &= \frac{1}{10} = 0.1\end{aligned}$$

Given:

$$\begin{aligned}\text{If } \ln x = 0, \text{ then } \ln y &= -3 \quad (\ln y - \text{intercept}) \\ \Rightarrow n &= -3\end{aligned}$$

Question 6

(a) Given function:

$$f(x) = (x - 2)(x + 3) + ax + b$$

Conditions:

$$f(2) = 8$$

$$(2-2)(2+3) + 2a + b = 8$$

$$b = 8 - 2a \quad (\text{Equation 1})$$

$$f(-3) = -12$$

$$(-3-2)(-3+3) + (-3)a + b = -12$$

$$(-5)(0) - 3a + b = -12$$

$$b = 3a - 12 \quad (\text{Equation 2})$$

Solving the equations:

$$\text{From Equation 1: } b = 8 - 2a$$

$$\text{Substitute into Equation 2: } 8 - 2a = 3a - 12$$

$$20 = 5a$$

$$a = 4$$

$$\text{Then from equation 1: } b = 8 - 2(4) = 0$$

$$\text{Finally: } a = 4 \quad \text{and} \quad b = 0$$

(b)

$$\frac{(x-2)(x+3) + 4x}{(x-2)(x+3)} = 1 + \frac{4x}{(x-2)(x+3)} \quad \left(\text{Quotient} + \frac{\text{Remainder}}{\text{Divisor}} \right)$$

$$\therefore \text{Remainder} = 4x$$

Question 7

$$x \frac{dy}{dx} = y + 1$$

$$\int \frac{1}{y+1} dy = \int \frac{1}{x} dx$$

$$\ln(y+1) = \ln x + c \quad [\text{General Solution}]$$

$$\text{If } y = 4; \quad x = 2$$

$$\ln(4+1) = \ln 2 + c$$

$$c = \ln\left(\frac{5}{2}\right)$$

$$\therefore \ln(y+1) = \ln x + \ln\left(\frac{5}{2}\right)$$

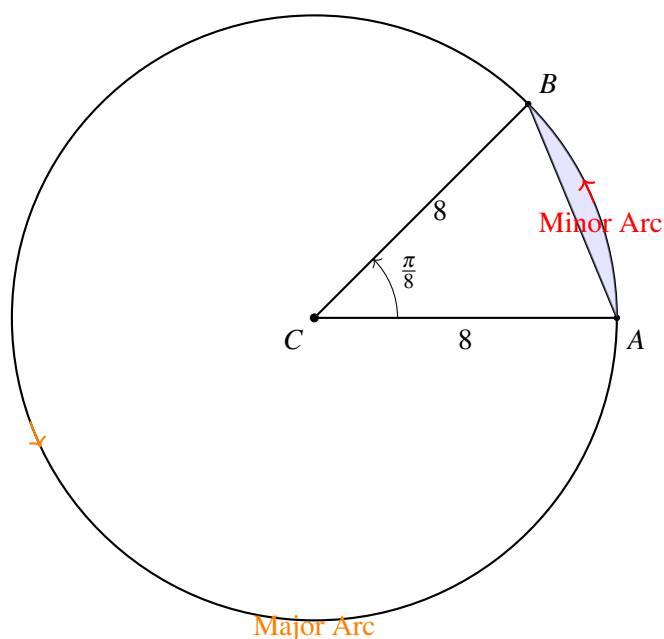
$$\ln(y+1) = \ln\left(\frac{5x}{2}\right)$$

$$(y+1) = \left(\frac{5x}{2}\right)$$

$$\therefore y = \frac{5}{2}x - 1 \quad [\text{Particular Solution or } (y = 2.5x - 1)]$$

Question 8

(a) Length of the major arc l :



$$\begin{aligned}l &= r\theta \\&= 8\left(2\pi - \frac{\pi}{8}\right) \\&= 15\pi \text{ cm}\end{aligned}$$

(b) Area of the minor segment A;

$$\begin{aligned}A &= \frac{1}{2}r^2(\theta - \sin \theta) \\&= \frac{1}{2}(8)^2\left[\frac{\pi}{8} - \sin \frac{\pi}{8}\right] \\&= 4\pi - 32 \sin \frac{\pi}{8} \\&= 0.32 \text{ cm}^2 \quad [2 \text{ s.f.}]\end{aligned}$$

Question 9

(a)

$$\begin{aligned}PQ &= \begin{pmatrix} -1 & 1 & 3 \\ 2 & -2 & -6 \\ -4 & 4 & 12 \end{pmatrix} \begin{pmatrix} 3 & -2 & 0 \\ -1 & 3 & 7 \\ 2 & -1 & 1 \end{pmatrix} \\&= \begin{pmatrix} -3-1+6 & 2+3-3 & 0+7+3 \\ 6+2-12 & -4-6+6 & 0-14-6 \\ -12-4+24 & 8+12-12 & 0+28+12 \end{pmatrix} \\&= \begin{pmatrix} 2 & 2 & 10 \\ -4 & -4 & -20 \\ 8 & 8 & 40 \end{pmatrix}\end{aligned}$$

(b) If a matrix does not have an inverse then: $\det = 0$ hence singular or non invertible matrix

$$\begin{aligned}\det &= -6 \begin{vmatrix} -6 & 0 \\ 10 & -4 \end{vmatrix} + 2 \begin{vmatrix} 4 & 0 \\ 8 & -4 \end{vmatrix} + 2 \begin{vmatrix} 4 & -6 \\ 8 & 10 \end{vmatrix} \\&= -6(24) + 2(-16) + 2(88) \\&= -144 - 32 + 176 \\&= 0 \quad (\text{Shown})\end{aligned}$$

Question 10

(a)

$$\begin{aligned}F &\propto W^2 \\F &= kW^2 \\165 &= k(830)^2 \\165 &= 688\,900k \\k &= \frac{33}{137\,780} \\F &= \frac{33}{137\,780}W^2\end{aligned}$$

(b) If $F = 32$;

$$\begin{aligned}132 &= \frac{33}{137\,780}W^2 \\W^2 &= \frac{132(137\,780)}{33} \\W &= \pm\sqrt{551\,120} \\W &= 742.3745685\dots \\W &= \pm 740 \quad [2 \text{ s.f.}]\end{aligned}$$

Question 11

(a)

$$\begin{aligned}e^{-1/2x} &= 3x - 1 \\ \text{Let } f(x) &= e^{-1/2x} - 3x + 1\end{aligned}$$

Then;

$$\begin{aligned}f(0) &= e^{(0)} - 3(0) + 1 = 2 > 0 \\f(1) &= e^{-1/2} - 3 + 1 \\&= -1.393\dots < 0\end{aligned}$$

Hence, change of sign indicates root between 0 and 1.

(b)

$$e^{-\frac{1}{2}x} = 3x - 1$$

$$-\frac{1}{2}x = \ln(3x - 1)$$

$$x = -2\ln(3x - 1)$$

Putting subscripts;

$$x_{n+1} = -2\ln(3x_n - 1) \quad [\text{Shown}]$$

(c)

$$x_1 = 0.5$$

$$x_2 = -2\ln[3(0.5) - 1]$$

$$= -2\ln(0.5)$$

$$= 1.386\dots$$

$$= 1.386 \quad [3 \text{ d.p}]$$

Question 12

$$\sqrt[3]{1 - \sqrt{3}i} = (1 - \sqrt{3}i)^{1/3}$$

$$\text{Let } z = 1 - \sqrt{3}i$$

$$r = \sqrt{1^2 + (-\sqrt{3})^2} = 2$$

$$\theta = \tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}$$

$$z = 2 \left[\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right) \right]$$

$$\begin{aligned} \therefore z^{1/3} &= 2^{1/3} \left[\cos\left(-\frac{\pi}{3} + 2k\pi\right) + i \sin\left(-\frac{\pi}{3} + 2k\pi\right) \right] \\ &= \sqrt[3]{2} \left[\cos\left(-\frac{\pi + 6k\pi}{9}\right) + i \sin\left(-\frac{\pi + 6k\pi}{9}\right) \right] \end{aligned}$$

For $k = 0$

$$\begin{aligned} z_1 &= \sqrt[3]{2} \left[\cos\left(-\frac{\pi}{9}\right) + i \sin\left(-\frac{\pi}{9}\right) \right] \\ &= 1,183938513 \dots - 0,4309 \dots i \\ &= 1,2 - 0.43i \quad [2 \text{ s.f}] \end{aligned}$$

For $k = 1$

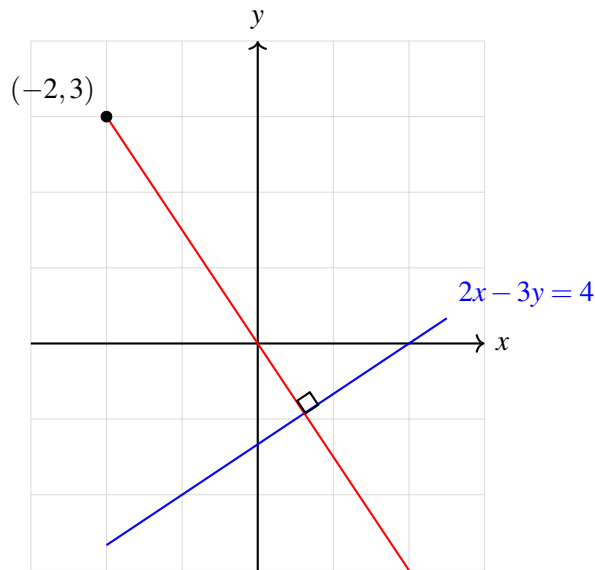
$$\begin{aligned} z_2 &= \sqrt[3]{2} \left[\cos\left(\frac{5\pi}{9}\right) + i \sin\left(\frac{5\pi}{9}\right) \right] \\ &= -0,218782 \dots + 1,24078 \dots i \\ &= -0.22 + 1.2i \quad [2 \text{ s.f}] \end{aligned}$$

For $k = -1$

$$\begin{aligned} z_3 &= \sqrt[3]{2} \left[\cos\left(-\frac{7\pi}{9}\right) + i \sin\left(-\frac{7\pi}{9}\right) \right] \\ &= -0,9651555 \dots - 0,809821640 \dots i \\ &= 0.97 - 0.81i \quad [2 \text{ s.f}] \end{aligned}$$

Question 13

(a) Sketch



$$2x - 3y = 4 \quad (1)$$

$$3y = 2x - 4 \quad (2)$$

$$y = \frac{2}{3}x - \frac{4}{3} \quad (3)$$

Slope (Gradient, m) of line l :

$$m = \frac{-1}{\frac{2}{3}} = \frac{-3}{2}$$

Equation of line l :

$$y - 3 = -\frac{3}{2}(x + 2) \quad (4)$$

$$y = -\frac{3}{2}x - 3 + 3 \quad (5)$$

$$y = -\frac{3}{2}x \quad (2) \quad (6)$$

Substitute equation (2) into (1):

$$2x - 3\left(-\frac{3}{2}x\right) = 4 \quad (7)$$

$$2x + \frac{9x}{2} = 4 \quad (8)$$

$$\frac{13x}{2} = 4 \quad (9)$$

$$x = \frac{4 \times 2}{13} = \frac{8}{13} \quad (10)$$

From equation (2):

$$y = -\frac{3}{2} \left(\frac{8}{13} \right) = \frac{-12}{13}$$

$$\therefore \text{Coordinates} = \left(\frac{8}{13}, -\frac{12}{13} \right) \quad (\text{Point P})$$

(b) Distance, d

$$\begin{aligned} d &= \sqrt{\left(-2 - \frac{8}{13}\right)^2 + \left(3 + \frac{12}{13}\right)^2} \\ &= \sqrt{\frac{289}{13}} \\ &= \frac{17\sqrt{13}}{13} \quad \text{units (exact form)} \end{aligned}$$

Question 14

(a)

$$\begin{aligned} -2 * 7 &= -2 + 7 + 1 \\ &= 6 \end{aligned}$$

(b) Closure property

Let $a, b \in \mathbb{Z}$. Then,

$$a * b = a + b + 1 \in \mathbb{Z}$$

Since integers are closed under addition and 1 is an integer, $a * b \in \mathbb{Z}$. Thus, closure holds true.

Associativity

$$\begin{aligned} (a * b) * c &= (a + b + 1) * c \\ &= a + b + 1 + c + 1 \\ &= a + b + c + 2 \end{aligned}$$

$$\begin{aligned}a * (b * c) &= a * (b + c + 1) \\&= a + b + c + 1 + 1 \\&= a + b + c + 2\end{aligned}$$

Associativity holds true since $(a * b) * c = a * (b * c)$.

Identity Element

We seek an element $e \in \mathbb{Z}$ such that for any $a \in \mathbb{Z}$:

$$a * e = a \quad \text{and} \quad e * a = a$$

So:

$$\begin{aligned}a * e &= a + e + 1 = a \\ \Rightarrow e + 1 &= 0 \\ \Rightarrow e &= -1\end{aligned}$$

Similarly:

$$\begin{aligned}e * a &= e + a + 1 = a \\ \Rightarrow e + 1 &= 0 \\ \Rightarrow e &= -1\end{aligned}$$

Thus, $e = -1$ is the identity element in $\mathbb{Z}$, hence identity exists.

Inverse Element

For each $a \in \mathbb{Z}$, we seek an element $b \in \mathbb{Z}$ such that:

$$\begin{aligned}a * b &= e = -1 \\ \Rightarrow a + b + 1 &= -1 \\ \Rightarrow b &= -2 - a \\ \therefore a^{-1} &= -a - 2 \\ &= b \in \mathbb{Z}\end{aligned}$$

Hence, inverse exists.

Commutativity:

$$a * b = a + b + 1$$

$$b * a = b + a + 1 = a + b + 1$$

Since $a * b = b * a$ hence abelian.

The set of integers $\mathbb{Z}$ with the binary operation $a * b = a + b + 1$ forms an **Abelian group**.

Question 15

(a)

$$f(f(2)) = 10 \quad (\text{Compound function})$$

$$f[3(2) + p] = 10$$

$$f(6 + p) = 10$$

$$3(6 + p) + p = 10$$

$$18 + 3p + p = 10$$

$$4p = -8$$

$$p = -2$$

Let $y = g(x)$:

$$y = q - 2x$$

Solve for x :

$$x = \frac{q - y}{2}$$

Thus, the inverse function is:

$$g^{-1}(x) = \frac{q - x}{2}$$

Given $g^{-1}(2) = 3$:

$$\frac{q - 2}{2} = 3$$

$$q - 2 = 6$$

$$q = 8$$

(b) Composite function $fg(x)$:

$$\begin{aligned}fg(x) &= f(8 - 2x) \\&= 3(8 - 2x) + (-2) \\&= 24 - 6x - 2 \\&= 22 - 6x\end{aligned}$$

Question 16

Let P_n be the statement:

$$\frac{5^{2n} + 5}{5} \text{ is divisible by 3}$$

Base Case ($n = 1$):

$$\begin{aligned}\frac{5^{2(1)} + 5}{5} &= \frac{25 + 5}{5} \\&= \frac{30}{5} \\&= 6 \quad (\text{which is divisible by 3 and also a multiple of 3})\end{aligned}$$

Thus, P_1 is true.

Inductive Hypothesis: Assume P_k is true for some $n = k$, i.e.:

$$\frac{5^{2k} + 5}{5} = 3d \quad \text{for some integer } d$$

which implies:

$$5^{2k} + 5 = 15d$$

$$5^{2k} = 15d - 5$$

Inductive Step ($n = k + 1$):

$$\begin{aligned}\frac{5^{2(k+1)} + 5}{5} &= \frac{5^{2k+2} + 5}{5} \\&= \frac{25 \cdot 5^{2k} + 5}{5} \\&= \frac{25(15d - 5) + 5}{5} \quad (\text{by inductive hypothesis}) \\&= \frac{375d - 125 + 5}{5} \\&= 75d - 24 \\&= 3(25d - 8) \quad (\text{Hence, } P_{k+1} \text{ is also true})\end{aligned}$$

Since P_1 , P_k and P_{k+1} hold true, hence by mathematical induction P_n is true, $\forall n \in \mathbb{Z}$.

Question 17

(a) Given:

$$f(x) = \frac{3x+14}{(x-2)(x^2+6)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+6}$$

Now, solving for coefficients A , B , and C :

If $x = 2$:

$$A = \frac{3(2)+14}{2^2+6} = \frac{20}{10} = 2$$

If $x = 0$:

$$\begin{aligned}\frac{14}{(-2)(6)} &= \frac{A}{-2} + \frac{C}{6} \Rightarrow -\frac{7}{6} = -\frac{2}{2} + \frac{C}{6} \\ -\frac{7}{6} &= -1 + \frac{C}{6} \Rightarrow \frac{C}{6} = \frac{-1}{6} \Rightarrow C = -1\end{aligned}$$

If $x = 1$:

$$\begin{aligned}\frac{17}{(-1)(7)} &= \frac{2}{-1} + \frac{B(1)-1}{7} \Rightarrow -\frac{17}{7} = -2 + \frac{B-1}{7} \\ -\frac{17}{7} + 2 &= \frac{B-1}{7} \Rightarrow -\frac{3}{7} = \frac{B-1}{7} \Rightarrow B = -2\end{aligned}$$

Thus, the partial fractions decomposition is:

$$f(x) = \frac{2}{x-2} + \frac{-2x-1}{x^2+6}$$

Alternative form:

$$f(x) = \frac{2}{x-2} - \frac{2x+1}{x^2+6}$$

(b)

$$\int_3^4 \left(\frac{3x+14}{(x-2)(x^2+6)} \right) dx = \int_3^4 \left(\frac{2}{x-2} + \frac{-2x-1}{x^2+6} \right) dx$$

$$\int_3^4 \left(\frac{2}{x-2} + \frac{-2x-1}{x^2+6} \right) dx = \int_3^4 \left(\frac{2}{x-2} + \frac{-2x}{x^2+6} + \frac{-1}{x^2+6} \right) dx$$

$$= \int_3^4 \left(\frac{2}{x-2} + \frac{-2x}{x^2+6} - \frac{1}{x^2+(\sqrt{6})^2} \right) dx$$

$$= \left[2 \ln|x-2| - \ln|x^2+6| - \frac{1}{\sqrt{6}} \tan^{-1} \left(\frac{x}{\sqrt{6}} \right) \right]_3^4$$

$$= \left[2 \ln 2 - \ln 22 - \frac{1}{\sqrt{6}} \tan^{-1} \left(\frac{4}{\sqrt{6}} \right) \right] - \left[-\ln 15 - \frac{1}{\sqrt{6}} \tan^{-1} \left(\frac{3}{\sqrt{6}} \right) \right]$$

$$= \ln 4 - \ln 22 - \frac{1}{\sqrt{6}} \tan^{-1} \left(\frac{2\sqrt{6}}{3} \right) + \ln 15 + \frac{1}{\sqrt{6}} \tan^{-1} \left(\frac{\sqrt{6}}{2} \right)$$

$$= \ln \left(\frac{4}{22} \times 15 \right) + \frac{1}{\sqrt{6}} \left[\tan^{-1} \left(\frac{\sqrt{6}}{2} \right) - \tan^{-1} \left(\frac{2\sqrt{6}}{3} \right) \right]$$

$$= \ln \left(\frac{30}{11} \right) + \frac{\sqrt{6}}{6} \left[\tan^{-1} \left(\frac{\sqrt{6}}{2} \right) - \tan^{-1} \left(\frac{2\sqrt{6}}{3} \right) \right] \quad (\text{exact form})$$

Standard Integral Formulas Used

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

$$\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

END OF SOLUTIONS

FEEL FREE TO CONTACT US FOR ANY ADJUSTMENTS, CLARIFICATIONS AND ASSISTANCE!

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